

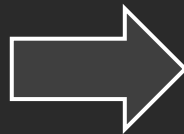
EM exercise 01.m

EM exercise 02.m

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1 # IT26102084 - Kuruvitaje.S.P.M
2 printf("IE1014 : Engineering Mathematics - I\n\n")
3 printf("***** Lab Sheet - 1.1 *****\n\n")
4
5 2^10          # a
6 sqrt(10)
7 exp(10)
8 log(10)
9
10 ceil(2.25)   # b
11 fix(2.25)
12 round(2.25)
13 floor(2.27)
14
15 sin(1)       # c
16 cosd(90)
17 asin(0.5)
18 atan(-1)
19
20 quotient = idivide(int32(9), int32(4)); # d
21 remainder = mod(9, 4);
22 printf("\nQuotient of 9 devided by 4 = %d\n", quotient)
23 printf("\nRemainder of 9 devided by 4 = %d\n\n", remainder)
24
25 M = [9.2, 40 ; 5^2, 3]; # e
26 printf("Determinant:\n")
27 disp(det(M))
28 printf("\nInverse:\n")
29 disp(inv(M))
30 printf("\nEigen:\n")
31 disp(eig(M))
32

```



octave:14> IE1014 : Engineering Mathematics - I

***** Lab Sheet - 1.1 *****

```

ans = 1024
ans = 3.1623
ans = 2.2026e+04
ans = 2.3026
ans = 3
ans = 2
ans = 2
ans = 0.8415
ans = 0
ans = 0.5236
ans = -0.7854

```

.....
Quotient of 9 devided by 4 = 2

Remainder of 9 devided by 4 = 1

Determinant:
-972.40Inverse:
-3.0852e-03 4.1135e-02
2.5710e-02 -9.4611e-03Eigen:
37.874
-25.674

```
1 # IT26102084 - Kuruvitaje.S.P.M
2 printf("IE1014 : Engineering Mathematics - I\n\n")
3 printf("***** Lab Sheet - 1.2 *****\n\n")
4
5 a = 10;
6 b = 5;
7
8 c1 = a + b ;
9 c2 = a - b ;
10 c3 = 10*a + 5*b ;
11 c4 = a / b ;
12 c5 = a ^ b ;
13 printf('c1=%.2f\n c2=%.2f\n c3=%.2f\n c4=%.2f\n c5=%.2f\n', c1, c2, c3, c4, c5)
14
15 # x^5 + 3x^4 + 2x^3 + 9x^2 + 4x = 0
16
17 poly = [1 ,3 ,2 ,9 ,4 ,0 ];
18 r = roots(poly);
19 printf("\n Roots of the polynomial are :\n\n");
20 disp(r)
21
```



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***** Lab Sheet - 1.2 *****

c1=15.00
c2=5.00
c3=125.00
c4=2.00
c5=100000.000000

Roots of the polynomial are :

-3.1453 + 0i
0.3047 + 1.6270i
0.3047 - 1.6270i
-0.4641 + 0i
0 + 0i